****

ACADEMIC YEAR : 2024 – 25

CLASS & BRANCH : II B.TECH, II SEM, ECE

SUBJECT NAME : **ELECTROMAGNETIC WAVES AND TRANSMISSION LINES (23A04407A)**

****

CO	COURSE OUTCOMES	K LEVEL
CO1	Apply the laws & theorems of electrostatic fields to solve the related problems.	K3
CO2	Gain proficiency in the analysis and application of magnetostatic laws and also analyze Maxwell's equations in different forms.	K4
CO3	Analyze electromagnetic wave propagation in different media.	K4
CO4	Explain the properties of transmission lines.	K3
CO5	Analyze the various impedance matching techniques and transmission line applications.	K4

QUESTION BANK

Essay Questions

Unit – I:

Q. NO.	QUESTION	CO	K LEVEL
1	(a) State and explain coulomb's law? Obtain an expression in vector form?	1	K2
	(b) A point charge, Q1 = 2μC is at (2, 3, 6) and another charge, Q2 = 5μC is at (0, 0, 0) in free space. Find the force on Q1 due to Q2.	1	K3
2	(a) State and explain Gauss Law.	1	K2
	(b) Determine an expression for the electric field E due to infinite line charge using Gauss Law.	1	K3
3	(a) Determine an expression for the electric field E due to infinite charge sheet using Gauss Law.	1	K3
	(b) If an electric field in free space is given by $E = a_x + 2a_y + 5a_z$ V/m, find the Electric flux density and $€_r = 4$.	1	K3
4	Derive two Maxwell's equations for electrostatic field.	1	K3
5	(a) Derive the Poisson and Laplace Equations and mention their Applications.	1	K3

	(b) Derive the relationship between Electric Field and Electric Potential.	1	K3
6	Define capacitance. Derive the capacitance of a parallel plate capacitor.	1	K3

Unit – II:

Q. NO.	QUESTION	CO	K LEVEL
1	(a) State and explain Biot-Savart Law.	2	K2
	(b) Derive the magnetic field intensity due to an Infinite line placed along z-axis at an observation point P on y axis.	2	K3
2	(a) State and explain the ampere's circuital law.	2	K2
	(b) State and explain Faraday's law of induction. And hence introduce the concept of time varying electromagnetic fields through this law.	2	K3
3	(a) Derive the expression for Energy density for steady magnetic field.	2	K3
	(b) Distinguish the scalar and vector magnetic potential.	2	K3
4	Describe the inconsistency in Ampere's law? How it is rectified by Maxwell?	2	K4
5	Derive the four Maxwell's equations both in integral and differential forms.	2	K3
6	(a) Write down the Maxwell's equations in word statement.	2	K4
	(b) A charge of 12 C has velocity of $5a_x + 2a_y - 3a_z$ m/s. Determine 'F' on the charge in the field of $B = 4a_x + 4a_y + 3a_z Wb/m^2$.	2	K3

Unit – III:

Q. NO.	QUESTION	CO	K LEVEL
1	(a) State and explain the poynting theorem and derive the expression for poynting vector.	3	K4
	(b) Derive the wave equation for uniform plane wave in terms of E.	3	K4
2	(a) Show that $\frac{E}{H} = \sqrt{\frac{\mu}{\varepsilon}}$ for a uniform plane wave.	3	K3
	(b) Explain and derive the skin depth and surface impedance.	3	K2

3	(a) Discuss the wave propagation in lossless dielectrics.	3	K4
	(b) Illustrate the concept of EM wave Polarization.	3	K3
4	(a) Illustrate the concept of reflection of plane wave at normal incidence and hence derive expressions for transmission coefficient and reflection coefficient.	3	K4
	(b) Find the depth of penetration, δ of an EM wave in copper at f = 50 kHz for copper, $\sigma = 5 \times 10^7$ mho/m, $\mu_r = 1$, $\epsilon_r = 1$.	3	K3

Unit – IV:

Q. NO.	QUESTION	CO	K LEVEL
1	(a) Derive the General solution of Transmission Line and explain its physical significance.	4	K3
	(b) Explain about phase velocity and group velocity.	4	K2
2	(a) Derive the attenuation constant α and phase constant β from the propagation constant.	4	K3
	(b) Derive the characteristic impedance Z_0 and propagation constant P from the initial equation of transmission line.	4	K3
3	(a) List the conditions of Lossless transmission line.	4	K3
	(b) Illustrate the concept of π Equivalent Circuit.	4	K3
4	(a) List the conditions of distortion less transmission line.	4	K3
	(b) Distinguish the various types of transmission lines.	4	K3

Unit – V:

Q. NO.	QUESTION	CO	K LEVEL
1	(a) Derive an expression for input impedance at any point on the transmission line.	5	K4
	(b) Illustrate the variation of Z_{in} of lossless transmission line when SC & OC with different lengths.	5	K4
2	(a) List the physical significance of Smith chart and its applications in transmission line.	5	K4
	(b) For a transmission line which is terminated in a normalized impedance Zn, VSWR = 2. Find the normalized impedance magnitude.	5	K4
3	(a) Show that infinite line is equivalent to a finite line terminated in Z0.	5	K3

	(b) Explain principle of impedance matching with quarter wave transformer.	5	K2
4	(a) Illustrates how UHF Lines act as Circuit Elements.	5	K3
	(b) A transmission line has a characteristic impedance of 300 ohms and terminated in a load ZL = (150+j 150) ohms. Find VSWR and reflection coefficient.	5	K4

Short Answer Questions

UNIT – I:

Q. NO.	QUESTION
1	Define Electric field and Electric flux density.
2	State and explain the Coulomb's law.
3	Define the convection and conduction currents.
4	List the limitations of Gauss's law.
5	What do you mean by conservative field?

UNIT – II:

Q. NO.	QUESTION
1	State Ampere's circuital law.
2	Explain briefly about Magnetic Flux Density.
3	State Biot-Savarts Law.
4	Write Maxwell's equations for time varying fields.
5	List the two Maxwell's equations for magnetostatic filed.

UNIT – III:

Q. NO.	QUESTION
1	Define polarization. Explain different types of polarization.
2	State and Express Poynting Theorem.

3	Compare Brewster Angle and critical angle?
4	Find the skin depth in copper if its conductivity $\sigma=5.8X10^7$ mho/m at 1 GHz.
5	List the properties of uniform plane wave.

UNIT – IV:

Q. NO.	QUESTION
1	Define the line parameters?
2	Define Characteristic impedance.
3	What is an infinite line?
4	Draw the T Equivalent Circuits.
5	Mention applications of the transmission lines.

UNIT – V:

Q. NO.	QUESTION
1	What is Impedance matching?
2	Define reflection coefficient.
3	What is smith chart? Mention its Applications.
4	Define the voltage standing wave ratio.
5	Define the stub.